SDOM Standard Inputs - Zonal Mode

This document describes every CSV input expected by SDOM when running in zonal mode (Network = AreaTransportationModelNetwork). Dummy CSV files matching this schema are provided in the same folder for reference. Wide files use the '@area_id@' tagging convention; row-oriented files add an 'area_id' column. For full details, see 'Zonal Inputs' in the User Guide.

formulations.csv

Selects modeling approach. For zonal runs, Network must be AreaTransportationModelNetwork.

Column	Type	Description
`Component`	string	Component name (Thermal, Hydro, Imports, Exports, Network).
`Formulation`	string	Formulation assigned to that component.
`Description`	string	Optional free-text description.

areas.csv

Defines the set of areas (zones) in the system. Required in zonal mode.

Column	Type	Description
`area_id`	string	Area identifier (primary key).
`description`	string	Free-text description of the area.

interconnections.csv

Topology of inter-area transmission lines. Required in zonal mode.

Column	Type	Description
`line_id`	string	Unique line identifier.
`from_area`	string	Origin area ID.
`to_area`	string	Destination area ID.

LineCap_FT.csv

Hourly directional line capacity for from_area -> to_area.

Column	Type	Description
`*Hour`	int	Hour index of the year.
	float	Line capacity in the from->to direction at each hour (MW). Non-negative.

LineCap_TF.csv

Hourly directional line capacity for to_area -> from_area.

Column	Type	Description
*Hour	int	Hour index of the year.
	float	Line capacity in the to->from direction at each hour (MW). Non-negative.

Load_hourly.csv

Hourly demand per area. Wide format with @area_id@ tagged headers.

Column	Type	Description
*Hour	int	Hour index of the year.
Load@@	float	Demand at each hour for the specified area (MWh).

CFSolar.csv

Hourly solar capacity factors per candidate site (plant-keyed; no @area@ tag).

Column	Type	Description
Hour	int	Hour index of the year.
	float	Capacity factor [0, 1] for each candidate solar site.

CapSolar.csv

Candidate solar sites with an area assignment. IDs must be globally unique across areas.

Column	Type	Description
sc_gid	string	Unique candidate site identifier (globally unique across areas).
area_id	string	Area the site belongs to.
capacity	float	Upper bound for installed capacity at the site (MW).
latitude	float	Latitude of the site (optional).
longitude	float	Longitude of the site (optional).
trans_cap_cost	float	Transmission interconnection capex (USD/kW).
CAPEX_M	float	Capital expenditure (USD/kW).
FOM_M	float	Fixed O&M; cost (USD/kW-yr).

CFWind.csv

Hourly wind capacity factors per candidate site (plant-keyed; no @area@ tag).

Column	Type	Description
Hour	int	Hour index of the year.
	float	Capacity factor [0, 1] for each candidate wind site.

CapWind.csv

Candidate wind sites with an area assignment (same schema as CapSolar.csv).

Column	Type	Description
`sc_gid`	string	Unique candidate site identifier (globally unique across areas).
`area_id`	string	Area the site belongs to.
`capacity`	float	Upper bound for installed capacity at the site (MW).
`latitude`	float	Latitude of the site (optional).
`longitude`	float	Longitude of the site (optional).
`trans_cap_cost`	float	Transmission interconnection capex (USD/kW).
`CAPEX_M`	float	Capital expenditure (USD/kW).
`FOM_M`	float	Fixed O&M; cost (USD/kW-yr).

Nucl_hourly.csv

Hourly nuclear generation per area. Wide format with @area_id@ tagged headers.

Column	Type	Description
`*Hour`	int	Hour index of the year.
`Nuclear@@`	float	Nuclear generation at each hour for the specified area (MWh).

lahy_hourly.csv

Hourly hydro time-series per area. Wide format with @area_id@ tagged headers.

Column	Type	Description
`*Hour`	int	Hour index of the year.
`LargeHydro@@`	float	Hydro profile or budget basis at each hour (MWh).

otre_hourly.csv

Hourly other-renewable generation per area. Wide format with @area_id@ tagged headers.

Column	Type	Description
`*Hour`	int	Hour index of the year.
`OtherRenewables@@`	float	Other-renewable generation at each hour (MWh).

StorageData.csv

Storage parameters per technology per area. Headers use @area_id@ tags (e.g., Li-Ion@A1@).

Column	Type	Description
`P_Capex`	float	Power-related capital expenditure (USD/kW).
`E_Capex`	float	Energy-related capital expenditure (USD/kWh).

Column	Type	Description
`Eff`	float	Round-trip efficiency (fraction).
`Min_Duration`	float/int	Minimum storage duration (h).
`Max_Duration`	float/int	Maximum storage duration (h).
`Max_P`	float/int	Maximum power capacity (kW).
`MaxCycles`	int	Maximum number of charge/discharge cycles.
`Coupled`	int (0/1)	1 if input and output power are coupled, else 0.
`FOM`	float	Fixed O&M; cost (USD/kW/year).
`VOM`	float	Variable O&M; cost (USD/MWh).
`Lifetime`	int	Expected system lifetime (years).
`CostRatio`	float	Cost allocation between input/output power (0.5 = equal split).

Data_BalancingUnits.csv

Thermal / balancing units with an area assignment. Plant_id must be globally unique.

Column	Type	Description
`Plant_id`	string	Unique plant or aggregation identifier (globally unique across areas).
`area_id`	string	Area the unit belongs to.
`MinCapacity`	float	Minimum installed capacity (MW).
`MaxCapacity`	float	Maximum installed capacity (MW).
`Lifetime`	int	Operational lifetime (years).
`Capex`	float	Capital expenditure (USD/kW).
`HeatRate`	float	Heat rate (MMBtu/MWh).
`FuelCost`	float	Fuel cost (USD/MMBtu).
`VOM`	float	Variable O&M; cost (USD/MWh).
`FOM`	float	Fixed O&M; cost (USD/kW).

scalars.csv

Global scalar parameters for the optimization (same as copperplate).

Column	Type	Description
`LifeTimeVRE`	int	Operational lifetime of VRE assets (years).
`GenMix_Target`	float	Target renewable generation share [0, 1].
`AlphaNuclear`	int (0/1)	Activate (1) or deactivate (0) nuclear.
`AlphaLargHy`	int (0/1)	Activate (1) or deactivate (0) large hydro.
`AlphaOtheRe`	int (0/1)	Activate (1) or deactivate (0) other renewables.
`r`	float	Discount / interest rate.
`EUE_max`	float	Maximum allowed Expected Unserved Energy.